SAMYAK DUGAD

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EDUCATION

Vishwakarma Institute of Information Technology (VIIT), Pune, India *September 2023 – May 2027*
B.Tech in Computer Engineering (Software Engineering) | GPA: 8.83 / 10

SKILLS

Languages: Python, C++, JavaScript
ML / AI: Scikit-Learn, TensorFlow, LangChain, RAG, LangGraph, Agentic AI, n8n, SHAP, Prophet, CTGAN
Systems / Cloud: AWS, Docker, Redis, PostgreSQL, MinIO, Nginx, Kubernetes
Full-Stack: FastAPI, Express/Node.js, React, Next.js, React Native, MongoDB, MySQL

PATENTS & RESEARCH

"A Real-Time Monitoring System for Elderly People"
Indian Patent Application (search 08/2026, then check the Publication No. column), published Feb 2026

- Contributed to the AI/analytics module for real-time anomaly detection (falls, abnormal vitals, prolonged inactivity) in a BLE-based wearable monitoring system.

"Adaptive Confidence-Driven Multispectral System for Low-Cost Plastic Waste Identification and Sorting"
Patent Filing in Progress

- Designed an embedded ML system using confidence-triggered adaptive multispectral scanning, reducing sensing operations by ~21–27% versus fixed full-spectrum acquisition.

"Evaluating the Utility-Privacy Trade-off of Synthetic Tabular Data Generation Under Real-Data Scarcity"
Zenedo Preprint

- Benchmarked Gaussian Copula and CTGAN generators across real-data scarcity levels using membership-inference and attribute-inference attacks on the Rossmann retail dataset to evaluate the utility-privacy trade-off.

"An Explainable Deep Learning Framework for Multi-Factor Demand Forecasting and Inventory Decision Support"
Working Paper

- Compared SARIMA, XGBoost, LSTM, GRU, and Transformer models on the M5 (Walmart) retail dataset with SHAP-based interpretability, and translated forecasts into safety-stock and reorder-point inventory policy.

WORK EXPERIENCE

Sankalp Computers and Systems Pvt. Ltd. — Internship *May 2026 – Oct 2026*
7K OTT Platform — Scrollable short-video streaming platform

- Designed and built a full production-grade OTT platform from a blank-slate architecture, supporting 1,000+ concurrent users.
- Engineered a VOD pipeline using MinIO object storage, a BullMQ job queue, and FFmpeg-based transcode workers for adaptive-bitrate encoding.
- Built a multi-layer Nginx caching system (HLS segment, image, and live-stream cache zones) with Cloudflare real-IP passthrough and keepalive upstream pooling, keeping cache hit rates high and end-to-end latency low.
- Built a modular Node.js/Express + Prisma/PostgreSQL backend (auth, membership, payments, content, live, CMS, notifications) with Redis and Meilisearch, two frontends (React/Vite admin panel, Expo/React Native mobile client), and a fully Dockerized multi-service deployment.

INDUSTRY PROJECTS

TCS — LLM Fine-Tuning Web UI [*Report*] *January 2026 – April 2026*
FastAPI, React/Vite, PyTorch, PEFT/LoRA

- Built an end-to-end LoRA fine-tuning platform covering dataset upload/validation, configurable-hyperparameter training, ROUGE-L/Exact-Match evaluation, and merged/adapter model export.
- Implemented live training and export status via Server-Sent Events (SSE), and designed the REST API architecture across model, dataset, training, and export lifecycles.

Alhaswise, Philippines — Venta AI: Sales & Inventory Analytics Platform [Report] *October 2025 – December 2025*

FastAPI, Streamlit, Prophet, LangChain

- Built a forecasting and inventory-optimization system (safety stock, reorder points, stockout projection) using Prophet time-series models with custom seasonality.
- Developed a LangChain SQL Agent-based chatbot translating natural language into secure T-SQL queries for real-time reporting.

Emmerson Technologies — BOM-to-DCS Architecture Diagram Generator [Report] *May 2025 – July 2025*

Python, Streamlit, SVG

- Built a tool that parses Excel Bills-of-Materials into automated DCS control-system architecture diagrams (cabinet/rack/tower layout, cable routing), eliminating manual diagram drafting.

TECHNICAL PROJECTS

Phishing Website Detection — End-to-End ML & MLOps [GitHub]

Docker, AWS ECR/EC2, CI/CD

- Built a MongoDB-sourced ETL pipeline and trained Decision Tree, Naive Bayes, and Logistic Regression models (~97% accuracy), assessed via accuracy, precision, recall, and ROC-AUC.
- Containerized and deployed the model via an automated CI/CD pipeline to AWS ECR & EC2.

AI-Powered Mortgage Eligibility System [GitHub]

RAG, FAISS, OpenAI, OCR

- Built a loan-eligibility pipeline validating borrower profiles against LTV, DTI, and RBI regulations, using a RAG pipeline (FAISS + OpenAI) over RBI/IRDAI regulatory data for eligibility reasoning.
- Automated borrower data intake with OCR + regex extraction across KYC documents, income proofs, bank statements, and property documents.

Hierarchical Movie Recommendation System — Unsupervised, Collaborative & Deep Learning [GitHub]

Python, PyTorch, Scikit-Learn, FAISS, UMAP

- Built a 3-milestone recommender on MovieLens-25M (162K users, 24K movies): K-Means + UMAP behavioral clustering (8 taste segments), SVD collaborative filtering, a content-based hybrid using 1,128-dim genome-tag vectors for cold-start handling, and a PyTorch Two-Tower neural retrieval model (62.2% validation accuracy on 4M interactions).
- Evaluated all models with a temporal train/val/test split to prevent leakage; SVD achieved Recall@10 = 5.14% (128× random baseline), and proposed a two-stage retrieval-ranking system design for 1M+ QPS serving.

AWS Cloud-Based ETL Pipeline

AWS S3, AWS Glue, Athena, QuickSight, Power BI

- Built an automated ETL pipeline using Amazon S3 and AWS Glue (Crawlers & ETL) for data cleaning, transformation, and feature engineering on sales data.
- Integrated Amazon Athena to query processed data and built interactive dashboards in QuickSight and Power BI for sales trend analysis.

OpenQC — EdTech Platform for Quantum Computing(openqc.in)

Next.js, Editor.js

- Collaborated on frontend development for a research/learning platform, integrating modern tools and design to support a better user journey.
- Implemented frontend features with Next.js and Editor.js, elevating UI/UX and optimizing the platform for learning and research.

LEADERSHIP & EXTRACURRICULAR

- **GFG (GeeksforGeeks) Club, VIIT —** Department Head: organized coding competitions and workshops.
- **SESA Departmental Club —** Event Management Head: planned and executed departmental events.
- **Winner, "Best Innovative Idea" —** Hack for Humanity Hackathon, for Unnatigram (rural business digital-growth platform).

CERTIFICATIONS & COURSES

- The Complete Full-Stack Web Development Bootcamp (Udemy)
- Complete Data Science, Machine Learning, DL, NLP Bootcamp (Udemy)
- AWS Cloud Technology Consultant (Coursera)